[Titel]

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August 11, 2026

Abstract

To be added

JEL Classification: C91, D91, O33

Keywords: algorithms; delegation; responsibility; punishment; experiment

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1 Introduction

Algorithms are increasingly involved in consequential decisions. Many of these decisions carry consequences that fall, at least in part, on others. Physicians rely on diagnostic systems when treating patients, firms let screening software rank job applicants, and financial advisors draw on algorithmic tools when managing client portfolios. In each case a human chooses whether to put the algorithm in charge. Algorithm use thus becomes a delegation decision and with that, two questions arise. Does delegating a decision to an algorithm change the responsibility that others assign to the person who delegated it? And does the prospect of being held responsible motivate delegation? This paper provides experimental evidence on both.

For delegation between humans, the experimental literature points in a clear direction. Delegation shifts responsibility. Principals who delegate an unpopular decision to a human intermediary are punished less for it, feel less responsible for it, and delegate strategically for precisely this reason (Bartling and Fischbacher, 2012; Hamman et al., 2010; Coffman, 2011; Oexl and Grossman, 2013). Recent evidence suggests that algorithms possibly inherit the intermediary's role. Delegation to an algorithm increases dishonesty by obscuring intent (Köbis et al., 2025), is used to extract surplus while shedding moral responsibility (Tontrup and Sprigman, 2025), and becomes more attractive when the moral stakes of the decision are real (Hüholt and Szech, 2026). Accordingly, decision-makers who face potential punishment should be drawn to the algorithm, and delegation should protect them when outcomes turn out badly. Yet no existing design varies whether punishment is possible, so whether anticipated punishment motivates delegation to an algorithm has not been tested directly.

We test both predictions in a preregistered online experiment with 322 participants.[1] A decision-maker, Player A, faces a prediction task (predicting a person's weight from a facial image and the person's height) with monetary consequences for a matched recipient, Player B. Player A has the option to either make the consequential prediction herself or delegate it to an algorithm that has been calibrated to perform equally well at the task. The environment we study, a prediction under genuine uncertainty made by a decision-maker with no material stake in the affected party's outcome, resembles the settings in which algorithmic delegation typically operates in practice. We then vary between subjects whether Player B can reduce Player A's earnings: In the *Punishment* condition, we elicit Player B's punishment for every combination of delegation choice and outcome, providing a within-subject test of whether delegation insulates Player A from punishment, conditional on the outcome. In the *No-Punishment* condition, no such punishment is possible. Comparing delegation rates across the two conditions turns our motivational question into a testable treatment comparison. With everything except the possibility of punishment held fixed, any difference in delegation is attributable to its prospect.

First, when punishment is possible, 42.5 percent of decision-makers delegate to the algorithm. When it is not, 59.3 percent do. Thus, we find that the prospect of punishment significantly

[1]The pre-registration (AsPredicted #133980) is available at https://aspredicted.org/hs9az8.pdf and reproduced in full in Appendix C.

reduces delegation shares by roughly 17 percentage points. Second, delegation offers no protection against punishment. Conditional on the outcome, the punishment of delegated and self-made decisions is statistically indistinguishable. Instead, punishment responds almost entirely to the outcome of the prediction, not to how it was made. Belief evidence suggests that decision-makers anticipate this pattern correctly.

Thus, the possibility of punishment moves the delegation decision even though no punishment attaches to delegation. We examine several candidate explanations. A first candidate is anticipated punishment. Player A may delegate less because she expects delegated decisions to be punished more harshly. Her elicited beliefs speak against this story. On average, her expected punishment is no higher for delegated than for self-made decisions, matching how Player B actually punishes. A second candidate operates via effort. Facing potential punishment, Player A may keep the decision in the hope of outperforming the algorithm. If so, non-delegators compared to delegators should improve on their earlier performance precisely when punishment is possible. Yet, actual performance improvement concentrates where punishment is impossible. What the data do show is a selection pattern. When punishment is possible, it is disproportionately the better-performing Player As who keep the decision for themselves. A preference for producing a good outcome oneself and a distaste for delegating uncertain decisions are both consistent with this pattern. However, both are treatment-independent in that they should operate in both conditions alike and thus predict no difference in delegation rates. Lastly, not delegating may carry signaling value. By keeping the decision, Player A may aim to convey to Player B that she intends to try especially hard on his behalf, whereas delegating could be perceived as shirking the decision. Since there is some inherent task uncertainty, sending this message does not require actually trying harder. The choice itself is the signal, so the motive is consistent with the absence of measurable extra effort among non-delegators. And unlike the preference explanations, it is inherently treatment-dependent. The signal is worth sending only when Player B holds sanctioning power, which could generate the delegation gap we observe.

Our contribution is twofold. First, we provide the first direct test of whether the prospect of punishment motivates delegation to an algorithm, the motive the responsibility-avoidance literature identifies for human intermediaries (Bartling and Fischbacher, 2012; Hamman et al., 2010; Coffman, 2011; Oexl and Grossman, 2013). Our environment departs from these canonical designs in three respects. The decision involves genuine uncertainty, so a bad outcome may reflect bad luck rather than bad intent. Its consequences fall entirely on the recipient, so there is no self-serving choice to police. And the intermediary is an algorithm rather than a person. In this environment, the motive operates with the opposite sign. Accountability disciplines algorithm use rather than fueling it, and it does so selectively among the decision-makers most likely to succeed. This rejection of the standard prediction in favor of the opposite direction is the paper's main empirical contribution, and we view it as a behavioral counterpart to the regulatory intuition that holding humans accountable for algorithm-mediated decisions disciplines their use of those algorithms (European Union, 2024). The reversal also adds a potential determinant of algorithm use that operates through the decision context rather than through the algorithm itself. Our results are consistent with decision-makers keeping the decision because delegating it may look like shirking to the person it affects, even when the algorithm performs equally well. Second, we document how affected parties assign responsibility for algorithmic delegation when the delegation choice is uninformative about ability. Punishment then tracks realized outcomes, and the insulation that delegation appeared to offer in earlier designs disappears.

Our design most closely relates to Feier et al. (2022), who find that decision-makers are held less responsible for outcomes delegated to an algorithm than for outcomes of their own actions. However, in their design each subject both delegates and evaluates the delegation decision, and under their session-level calibration of the algorithm, delegation is informative about and primarily driven by self-assessed relative ability. Subjects may then sanction perceived overconfidence rather than the delegation itself. Once these channels are closed, as in our design, the punishment differential disappears. Their treatments vary the type of delegate under a fixed punishment regime, whereas we hold the delegate fixed and vary the punishment regime. The two studies thus answer complementary questions, and ours isolates the role of the prospect of punishment.

The remainder of the paper is organized as follows. Section 2 relates the study to the literature. Section 3 presents the experimental design and hypotheses. Section 4 reports the results and examines candidate mechanisms. Section 5 concludes.

2 Related Literature

Responsibility avoidance through delegation

Delegation has long been studied as an efficiency-enhancing institutional arrangement with principals delegating to agents who have better information, lower opportunity costs, or specialized skills. Alongside these efficiency motives, the literature has identified a range of strategic motives for delegation, e.g. as a commitment device that alters the equilibrium (Schelling, 1960; Fershtman and Gneezy, 2001), as a way to motivate the agent (Aghion and Tirole, 1997), or as a means of shifting responsibility for the outcome away from the principal, i.e. principals may delegate not because the agent is more competent or cheaper, but because doing so shifts responsibility for the resulting outcome away from themselves — both in their own eyes (felt responsibility for the outcome) and in the eyes of others (the blame others attribute to them). The political-science and public-choice literatures have long recognized that delegation can serve as a strategy for shifting blame onto an intermediary (Weaver, 1986; Fiorina, 1986).

Experimental economics has established the same motive in incentivized settings, primarily using modified dictator games. Bartling and Fischbacher (2012) provide the canonical experimental evidence. In their design, a dictator may either choose between a fair and an unfair allocation of an endowment or instead delegate this choice to another person whose monetary incentives are aligned with the dictator's. A recipient, who is adversely affected by the unfair allocation, may then assign costly punishment to the dictator and the delegee. They find that through delegation punishment is effectively shifted from the dictator to the intermediary, and the prospect of avoiding punishment constitutes a strong motive for the delegation decision itself. Coffman (2011) reinterprets the mechanism behind this punishment reduction. He finds that the principal can no longer avoid punishment when she directly interacts with the receiver despite the intermediary's presence, suggesting that what reduces punishment is the absence of direct interaction with the affected party rather than any transfer of blame to a deciding intermediary. This holds even when the intermediary remains completely passive. Oexl and Grossman (2013) sharpen this reading further by modifying the canonical game so that delegation necessarily eliminates the fair option. The principal nevertheless escapes punishment, with the intermediary actively choosing between two unfair allocations after delegation. Finally, both Bartling and Fischbacher (2012) (with a die roll) and Oexl and Grossman (2013) (with an intermediary-triggered random draw) include treatments that substitute a randomization device for the human intermediary. In both, the principal still shifts some punishment to the device, though less than under delegation to a deciding human. Consistent with this, in Bartling and Fischbacher (2012) fewer dictators delegate to the die than delegate to a human delegee in the corresponding treatment. Two components of responsibility shifting thus emerge — the absence of direct interaction between principal and recipient drives the reduction in principal punishment, while the act of choosing, rather than any contribution to outcome unfairness, is what transfers blame to the intermediary.

The motive is not confined to settings with a punishment stage. Hamman et al. (2010) extend this picture in a setting without punishment opportunity. Still, principals' transfers to recipients fall sharply when they delegate, indicating that responsibility-shifting can operate through felt responsibility or moral wiggle room rather than through the avoidance of punishment. Qualitative self-reported responsibility and recipient-side blame attributions are consistent with this reading with dictators feeling less responsible when they delegate, and recipients placing less blame on principals than on intermediaries.

Beyond the canonical dictator-game setting, the responsibility-shifting motive, and the strategic use of delegation it gives rise to, has been documented across a range of domains. In incentivized ultimatum bargaining, proposers using a hired delegate extract larger surpluses (Fershtman and Gneezy, 2001). In sequential voting, non-pivotal voters are punished less than pivotal ones for the same group outcome and voters strategically vote against their preferred allocation to avoid being pivotal (Bartling et al., 2015). For acts of deception, principals pay an agent to lie on their behalf (Erat, 2013) and are more willing to lie through a delegate than directly (Gawn and Innes, 2019; Kandul and Kirchkamp, 2018). In policy-making, observers rate delegating legislatures as less blameworthy than legislatures who decide directly, even when the delegate is powerless to change the outcome (Hill, 2015). Freer et al. (2024) isolate the motive in financial decision-making, finding that a substantial fraction of investors delegate even purely luck-based lottery choices, where alternative motives such as decision costs, performance-chasing, or risk tolerance cannot rationalize the choice.

In sum, the canonical literature documents that in settings with self-interested choice, delegation reduces both the principal's punishment and her felt responsibility, in turn motivating delegation. Our setting departs from this canonical design along three dimensions. First, the intermediary is algorithmic rather than human. Second, the outcome falls entirely on a passive recipient rather than partly on the principal herself. Third, the underlying decision involves genuine predictive uncertainty rather than a known fair–unfair choice, removing the transparent self-serving intent that punishment sanctions in the canonical designs. Whether the responsibility-shifting motive retains its force under these conditions is the question we take up.

Delegation to algorithms

A large and rapidly growing empirical literature studies when and why people are willing to use algorithmic decision aids. Dietvorst et al. (2015) introduce the term *algorithm aversion* — the finding that people prefer their own (or other humans') judgement over algorithmic predictions, even after observing that algorithms outperform human decision-makers.[2] In contrast, Logg et al. (2019) document conditions under which subjects show *appreciation* of algorithmic advice over human advice. The picture that emerges is therefore one of substantial heterogeneity in algorithm preferences across tasks, contexts, and elicitation methods. Burton et al. (2020), Mahmud et al. (2022), Chugunova and Sele (2022), Qin et al. (2025) and Irlenbusch (2026) provide systematic reviews of this literature and document a large array of moderators for when and why algorithms are used in decision-making.

For example, on the algorithm side, algorithm aversion is stronger when the algorithm cannot be observed to learn from its mistakes (Berger et al., 2021), and when the algorithm remains a black box to its users (Litterscheidt and Streich, 2020). Evidence about relative performance gathered through experience and feedback reduces aversion (Filiz et al., 2021), whereas explanations of how the algorithm works leave acceptance unchanged (Dargnies et al., 2026). On the task side, algorithm aversion rises with the degree of irreducible uncertainty (Dietvorst and Bharti, 2020), with the perceived subjectivity of the task (Castelo et al., 2019), while reliance on algorithmic advice increases with task difficulty (Bogert et al., 2021) and under time pressure (Jung and Seiter, 2021). On the control side, people use imperfect algorithms far more readily when they retain even a minimal ability to modify the output (Dietvorst et al., 2018). Likewise, experimental firms delegate pricing to an algorithm more often when they can override its decisions (Normann et al., 2025). Part of algorithm aversion may thus reflect a reluctance to relinquish decision authority as such rather than distrust of algorithms (Ivanova-Stenzel and Tolksdorf, 2025), and performance feedback raises reliance on algorithms without eliminating

[2] Interest in the comparison between human and statistical judgement is long-standing — see Meehl (1954) and Dawes (1979) for the classic demonstrations that simple statistical rules outperform clinical judgement and are resisted nonetheless.

the desire for personal control (Ivanova-Stenzel and Tolksdorf, 2024).

To do: Supplement with more recent evidence; Explain that while context, task, etc. may matter for whether shifting responsibility is possible and this motive may impact demand for algorithms, level-effects of a pure preference for algorithms are constant across treatments.

Responsibility perceptions of decisions involving algorithms

Within this body of work, a smaller subset focuses on algorithm use preferences in settings whose consequences fall partly or wholly on someone other than the decision-maker, sometimes called the *moral domain* (Gogoll and Uhl, 2018). Much of this work concludes that aversion to algorithms is amplified once third-party welfare is at stake. Gogoll and Uhl (2018) document this in an incentivized calculation task with performance uncertainty, Bigman and Gray (2018) in vignettes across high-stakes moral domains (driving, legal, medical, military), and Jauernig et al. (2022) identify *moral discretion* as the human capacity that people specifically value in such settings.[3]

Once decisions affect a third party, the responsibility-shifting motive established for human delegation becomes relevant for algorithmic delegation as well — and with it the question of how observers apportion blame when an algorithm is involved in the decision. [Comment: In principle, felt responsibility may be reducable by delegating also in decisions that only affect oneself.] A first strand of evidence, predominantly vignette-based and unincentivized, examines how third parties attribute responsibility and blame to human actors involved with algorithms. In the medical domain, Pezzo and Pezzo (2006) document that a physician who follows a (bad) recommendation from a computerised decision aid is faulted less than one who errs unaided. Tobia et al. (2021) extend the discussion to liability law, where physicians who follow standard-care AI recommendations face reduced legal exposure. The authors argue that this constitutes an incentive to follow such recommendations specifically because doing so shields physicians from blame. In hiring, algorithmic discrimination triggers less moral outrage than identical human discrimination (Bigman et al., 2023). In the workplace setting, Naquin and Kurtzberg (2004) find that observers attribute less responsibility to a company when an accident involves technology. Across multiple moral domains (driving, legal, medical, military), Bigman and Gray (2018) document a robust "algorithmic outrage asymmetry", by which moral violations by algorithms produce less outrage than identical violations by humans. Shank et al. (2019) consider moral violations produced by human-algorithm pairs and find that humans who monitor algorithms are blamed less than humans acting alone or humans receiving algorithm recommendations, suggesting that responsibility attributions are sensitive to the precise division of authority between human and machine.

To do: Supplement/replace with more recent evidence.

[3]For recent surveys of this area see Bonnefon et al. (2024) and, with a legal-policy focus, Sunstein and Gaffe (2024).

Evidence using incentivized laboratory experiments is younger and smaller. Kirchkamp and Strobel (2019) compare a modified dictator game in which the dictator either acts alone or makes the decision jointly with a computer. They find no significant difference in the dictator's self-felt responsibility between human-computer and human-human teams, although a tendency towards more selfish allocations under shared decision-making with humans. Maasland and Weißmüller (2022) find in a computer-aided HR setting that algorithm aversion *dominates* blame avoidance, i.e. the prospect of using an algorithm to delegate an unpleasant HR decision (a dismissal) does not overcome the underlying aversion to algorithms in this domain.

A more recent strand documents the opposite pattern, in which algorithmic intermediaries serve as moral cover. Chevrier and Teixeira (2024) extend the blame chain to include the programmer behind the algorithm. In their design, AI programmers and the delegator both escape punishment for inegalitarian AI-mediated allocations, and the resulting "moral wiggle room" is exploited. Tontrup and Sprigman (2025) examine delegation of a dictator-game allocation to ChatGPT and find that delegators transfer less to the recipient than non-delegators, indicating that AI delegation is used strategically to extract surplus while shedding moral responsibility, similarly to delegation to other humans in the canonical literature. Hüholt and Szech (2026) use an incentivized trolley-style donation choice between two real charitable allocations and find that subjects delegate the moral decision to an AI significantly more often than to a human counterpart. Across these three different settings, the AI-as-blame-shield motive is shared, and the direction of the effect on delegation is uniformly positive — in contrast to Maasland and Weißmüller (2022), where algorithm aversion dominates the responsibility-shifting motive in the same broad domain.

The most closely related study is Feier et al. (2022), who study delegation to an algorithm in an incentivized laboratory experiment. In their design, to determine a third party's payoff subjects have the option to either rely on their own performance in a Raven-style logic puzzle or, depending on the treatment, delegate to another human or an algorithm. The subject's own payoff is unaffected by this choice. Third parties may then reward or punish the principal. The headline finding is that decision-makers are held *less* responsible for decisions delegated to an algorithm than for decisions delegated to another human, conditional on outcome. Interestingly, in their experiment, punishment is only reduced significantly when delegating to an algorithm, but not to another human, compared to self-made decisions. Thus, algorithm-delegation, but not human-delegation, shields the principal from blame for bad outcomes.

Our design extends Feier et al. (2022) along three dimensions. First, where their algorithm is calibrated at the session level so that the delegation decision reflects beliefs about own ability *relative* to that distribution, in turn constituting a signal value about Player A's self-perception that may contaminate downstream punishment as a measure of responsibility shifting per se, we calibrate the algorithm at the individual level, removing this relative-performance signal of delegation by construction.[4] Second, where the same subject acts as both delegator and

[4]Indeed, their own analysis shows that the propensity to delegate is driven primarily by self-assessed errors in the logic task, not by the type of delegate. Punishment of a non-delegator may then partly sanction perceived

evaluator in their design, potentially introducing self-justification, hypocrisy aversion, and similar motives into the punishment decision, we separate the delegator and evaluator roles across disjoint subject pools.[5] Third, their design does not include a treatment in which punishment is ruled out by construction, so that delegation rates between the two treatments may be different not because of differences in anticipated punishment, but more general preferences for algorithms. Thus, the motivational question of whether the prospect of punishment itself drives delegation is not identified. We add this manipulation directly, isolating the role of anticipated punishment in the delegation decision itself.

Thus, our experiment is, to our knowledge, the first to provide a clean test of blame-shifting onto an algorithm in a setting with inherent uncertainty, controlling for relative-performance concerns. It is also the first to vary the punishment possibility for an algorithmic delegate, and hence to test whether, in such a setting the prospect of punishment itself motivates the delegation decision.

3 Experimental Design

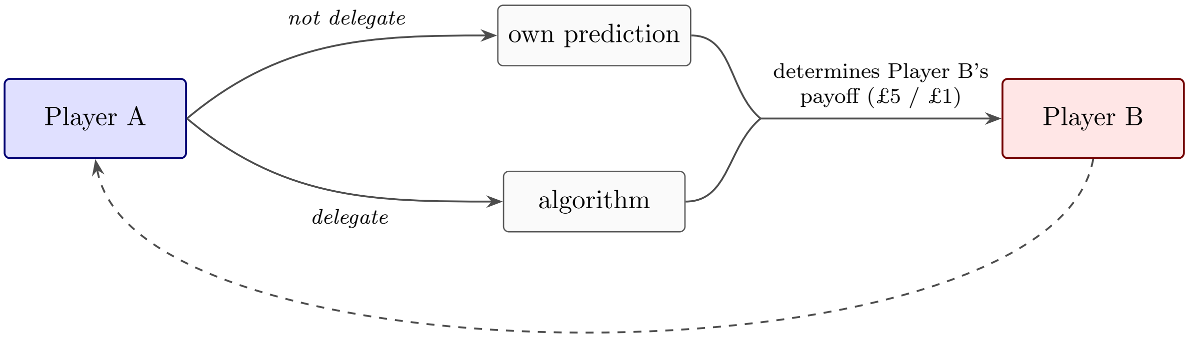
We ran an online experiment via the British platform Prolific (https://www.prolific.com) with UK participants. The experiment was programmed in oTree (Chen et al., 2016) and eligibility followed Prolific's standard quality screens, a minimum 95% prior approval rate, a fluent-English filter, and a desktop-only restriction (no mobile or tablet) that kept the screen layout consistent across subjects.

The design combines a between-subject manipulation of whether costly punishment is possible with a within-subject elicitation of conditional punishment behavior. Within each condition, participants assumed one of two roles. Player A decides whether to make a payoff-relevant prediction for Player B herself, or to delegate the prediction to an algorithm calibrated to match her own measured performance. Player B is the matched recipient whose payoff is fully determined by the outcome of the prediction, whether Player A makes it herself or delegates it to the algorithm. In the *Punishment* condition Player B may subsequently punish Player A, conditional on both the delegation decision and the realized outcome, while in the *No-Punishment* condition no punishment is possible. Figure 1 summarizes the basic setup.

overconfidence rather than responsibility-taking per se.

[5] For example, in their data, the evaluator's own delegation decision is significantly correlated with the evaluation gap between delegated and non-delegated decisions after good outcomes ($r = -0.34$ in the machine treatment; own computations from their published data)

Figure 1: Overview of the experimental conditions



***Punishment* condition:** Player B may reduce Player A's payoff by up to £2, conditional on delegation choice and outcome
***No-Punishment* condition:** no punishment option

Procedure

Player A

After providing informed consent, Player A completed ten incentivized rounds of a visual prediction task. Specifically, in each round, she predicted a person's weight from a facial image and the person's height. She could enter predictions via a slider or type them directly in kilograms, pounds, or stones-and-pounds.[6] No feedback was provided between rounds. At the end of the experiment, one of the ten rounds was randomly selected for payment. Player A earned £4 if her prediction in that round was within 10 lbs of the true weight and £2 otherwise. These initial rounds served three purposes simultaneously: First, they established a stake from which any later punishment could be deducted. Second, they familiarized Player A with the task. Third, they generated the performance data needed to calibrate the algorithm.

The choice of a visual prediction task is not incidental. Algorithm preferences may depend sensitively on the algorithm's perceived relative competence (Qin et al., 2025; Logg et al., 2019). Our design neutralizes this concern by calibrating the algorithm to match each subject's individual performance. The visual nature of the task lends credibility to this equal-performance claim, as people have everyday familiarity with judging others' build and appearance, and algorithmic error is plausible.[7] In contrast, subjects may be skeptical that an algorithm could ever fail in purely mathematical tasks (Gogoll and Uhl, 2018; Feier et al., 2022). Finally, we chose a prediction task involving genuine uncertainty rather than a setting with an obvious self-serving motive (as in classic dictator-game variations).[8] This reflects the nature of many real-world applications of algorithms, in which outcomes are inherently uncertain, the decision-maker has no direct material interest in the affected party's outcome, and performance is evaluated ex post.

[6]All individuals depicted in the images had explicitly consented to the use of their photographs for research purposes.
[7]Predicting body weight from an image has been used in earlier experiments (Logg et al., 2019).
[8]With no self-serving option available, an intention-based punishment motive should have less bite than in those settings.

After the ten rounds, we randomly paired Player A with another participant, Player B, who had not completed the prediction task himself. We informed Player A that her final prediction would not affect her own payoff but would almost entirely determine Player B's. A prediction within 10 lbs of the true weight would give Player B a high payoff (£5), and any less accurate prediction a low payoff (£1).[9] Player A then learned that she could either make the prediction herself or delegate it to an algorithm. She was also told that, based on her first ten predictions, the algorithm would have performed equally well.

We calibrated the algorithm to mimic Player A's own performance over the first ten rounds. Concretely, if Player A delegated the final prediction to the algorithm, the algorithm was equivalent to drawing a Bernoulli outcome with success probability equal to Player A's empirical accuracy rate over her first ten rounds. For example, a subject who scored 3 out of 10 in the initial rounds thus delegates to an algorithm that produces the high payoff for Player B with 30% probability.

Fixing the algorithm's success probability to each subject's own accuracy makes the equal-performance claim hold subject by subject, not merely on average across a pool of participants. This matters because beliefs about relative performance are otherwise likely a dominant driver of delegation.[10] For example, under the session-level calibration of Feier et al. (2022), the delegation choice is largely driven by each subject's sense of her ability relative to the algorithm. Matching the algorithm to each subject neutralizes this motive, and making this performance parity common knowledge should limit Player B from reading delegation as a signal of Player A's believed relative competence. This narrows a signal-extraction channel that may otherwise contribute to Player B's punishment.

In the *Punishment* condition, we further informed Player A that Player B could impose (costly) punishment of up to £2, and could condition punishment on both her delegation decision and the realized outcome. In the *No-Punishment* condition, Player B had no such punishment option. The two delegation options (own decision / algorithm) appeared in random order across subjects. We attached no monetary cost to delegation. This brings us closer to Player A's intrinsic preference over delegation in the *No-Punishment* condition, since with any positive cost, the responsibility-shifting motive would need to outweigh a price effect. Thus, costless delegation is also standard in the literature (Bartling and Fischbacher, 2012; Coffman, 2011; Feier et al., 2022). If Player A chose to make the prediction herself, she entered her final prediction on the next screen. If she delegated instead, this screen was skipped.

We then asked Player A to report her beliefs about Player B's punishment in each of the four scenarios resulting from the combination of delegation decision (yes/no) and prediction outcome (high/low payoff for Player B). We elicited beliefs on the same scale as actual punishment

[9]Since the final prediction determines only Player B's payoff, Player A has no financial interest in Player B receiving the low payoff. As long as a better outcome for Player B weakly reduces Player A's expected punishment in the *Punishment* condition, the incentives of the two players can be considered aligned.

[10]This motive would not inherently bias the treatment-effect comparison, since it should be balanced across the *Punishment* and *No-Punishment* conditions, but it might otherwise dominate the delegation decision and may obscure the responsibility-shifting motive of interest.

(£0–£2 in £0.10 increments) and presented the four scenarios in block-wise random order. We chose not to incentivize belief elicitation for three reasons. First, only two of the four scenarios remain payoff-relevant after the delegation decision, making the others purely hypothetical. Second, an incentivized mechanism could induce hedging. Third, an incentive-compatible scoring rule across four scenarios would have substantially increased task complexity. Because we are nonetheless interested in comparing punishment beliefs to actual punishment, we included an attention check on the belief-elicitation screen. In the *No-Punishment* condition, where no punishment was possible, we elicited beliefs about hypothetical punishment to keep the elicitation identical across conditions.

We then asked Player A to assess her own performance in the first ten rounds by assigning a probability mass (summing to 100%) to each possible score from 0 to 10.[11] We elicited this belief after the delegation decision, and after Player A had been told that the algorithm would perform equally well. While learning about algorithm performance may lead her to revise her beliefs about her own performance, it is precisely this post-update belief that is the analytically relevant input for any subsequent behavior. We also asked Player A to estimate the distribution of scores in the population, assigning to each score from 0 to 10 the share of participants she expected to have achieved it. Combined with her belief about her own score, this locates where she places herself in the distribution and yields a measure of perceived ability relative to other participants. One score was drawn at random at the end of the experiment, and Player A earned £0.30 if her estimate for it fell within five percentage points of the realized population share.

Finally, we elicited risk preferences using an incentivized multiple price list in the style of Holt and Laury (2002) and administered a short questionnaire on socio-demographic characteristics and technology affinity, before presenting Player A with her preliminary payoff.

Player B

After providing informed consent, Player B learned that he had been randomly matched with a Player A, and we introduced him to the task Player A had completed. He learned that his own payoff would be determined by the outcome of Player A's final prediction - £5 if it fell within 10 lbs of the true weight and £1 otherwise - while Player A's payoff depended solely on her performance in the initial ten rounds. Crucially, we also told Player B that Player A had the option to delegate the final prediction to an algorithm that performs as well as Player A had perfromed in the earlier rounds, and that Player A was aware of this, making it common knowledge.

In the *Punishment* condition, Player B could choose to reduce Player A's payoff by up to £2

[11] The elicitation was incentivized. Player A earned an additional £0.30 with the probability she had assigned to her actual score. While this linear scoring rule is not proper as its payoff-maximizing report places all mass on the single most likely score, irrespective of risk attitude, in practice most subjects spread probability mass across several scores.

in £0.10 increments. We focus on punishment rather than reward because the responsibility-attribution literature has predominantly operationalized blame through punishment (Bartling and Fischbacher, 2012; Coffman, 2011; Steffel et al., 2016). Imposing strictly positive punishment required Player B to forgo £0.10 of his own earnings, ensuring that punishment was costly to the punisher. We elicited punishment decisions for all four combinations of delegation (yes/no) and outcome (high/low payoff for Player B), implementing only the decision matching the realized combination. Because the punishment decision is the central outcome variable for Player B, we included an attention check on the punishment-elicitation screen. In the *No-Punishment* condition, where actual punishment was not possible, we elicited hypothetical punishment decisions on the same scale to keep the experimental flow comparable across conditions. Player B's perceptions of task difficulty, his risk preferences, and a short questionnaire followed, before we presented him with his final payoff.

After Player B's decisions were recorded, we calculated Player A's final payoff and paid all participants on the day of participation.

Hypotheses and identification

Our design tests two main hypotheses, both grounded in the literature on responsibility avoidance through delegation to human intermediaries (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013; Hamman et al., 2010; Hill, 2015; Steffel et al., 2016) and on responsibility attributions to algorithmic decision-makers (Feier et al., 2022; Kirchkamp and Strobel, 2019; Gogoll and Uhl, 2018). These strands of the literature mostly focus on what Gogoll and Uhl (2018) have coined the *moral* domain. Although the underlying prediction is not itself moral, Player A's choice imposes monetary consequences on a matched recipient, and a poor prediction causes real monetary harm. In this respect, our setting shares the defining feature of the settings this literature studies.

H1 (Delegation): The introduction of potential punishment increases the likelihood that Player A delegates the final prediction to the algorithm.

H1 is grounded in empirical evidence that responsibility avoidance shapes delegation decisions involving human intermediaries. Bartling and Fischbacher (2012) also show that subjects are more likely to delegate morally consequential decisions even to a die roll, an early form of delegation to a non-intentional agent, suggesting that responsibility avoidance persists when the agent lacks intentionality. Whether the same logic extends to *algorithmic* delegates, to decisions under genuine uncertainty, and to settings in which the outcome falls entirely on the recipient is less explored. Feier et al. (2022) report that delegation rates are similar for human and algorithmic intermediaries, suggesting that the kind of intermediary may matter less than the mere availability of one. Their design, however, does not directly compare delegation rates across punishment regimes, leaving open the question H1 addresses. H1 is identified by the between-subject comparison of Player A's delegation rate across the *Punishment* and *No-Punishment* conditions, which exploits the punishment manipulation as the only between-subject difference and holds fixed every other feature of the decision environment.

H2 (Punishment): Conditional on a low-payoff outcome, Player B punishes Player A less when she delegated the final prediction to the algorithm than when she made it herself.

H2 mirrors H1 on the punishment side. Bartling and Fischbacher (2012) find that delegation to a randomization device still allows decision-makers to avoid some punishment, although less so than delegation to another human. Feier et al. (2022) find that decision-makers are rewarded less when a negative outcome is directly attributable to their own action, compared to when it results from an algorithm to which they delegated. Our study differs from theirs in three respects. First, the task differs. Second, the roles of Player A and Player B are separated across disjoint subject pools. Third, their design does not control for relative performance at the individual level. Under their calibration, the delegation decision carries a strong signal about Player A's beliefs concerning her own ability relative to the algorithm. A failure to delegate may then be punished as overconfidence rather than for assuming responsibility, confounding inference on responsibility-shifting through pure intermediation with an inference about ability. H2 is identified by the within-subject comparison across the four (delegation, outcome) scenarios of Player B's strategy-method punishment, which isolates the effect of delegation on punishment, holding fixed both the realized consequences for Player B and all between-subject heterogeneity in Player B's punishment proclivity.

4 Results

Data collection took place between February and June 2023 through the British online platform Prolific, with participation restricted to UK residents. A total of 322 subjects were recruited (161 assigned to Player A, 161 to Player B).[12] On average, Player A spent 12 minutes completing the experiment and earned £3.63, while Player B spent approximately 8 minutes and earned £2.40. Treatment randomization was effective, as demonstrated by the balance across observable characteristics (Appendix A.1, Table 4).

In the following, we first test the two main hypotheses derived from our preregistered research questions. The subsequent analyses serve to understand the mechanism behind these results. They are exploratory in nature, though several of them were outlined as secondary analyses in the pre-registration.[13]

[12] We preregistered the collection of data from 320 participants (80 subjects per role per treatment). Due to attrition and subsequent re-matching we recruited an extra pair in the *No-Punishment* condition.

[13] Unless stated otherwise, all tests are two-sided. We describe results as significant if $p < 0.05$ and as weakly significant if $0.05 \leq p < 0.10$. Because the delegation decision is a binary outcome compared between conditions, delegation shares are tested using Pearson Chi-squared tests, with Fisher's exact tests yielding the same conclusions throughout. Because punishment is elicited within-subject through the strategy method,

Player A – Delegation behavior

We begin by examining the delegation behavior of Player A. In the *Punishment* condition, where Player B can punish, 42.5% of Player As delegated the final prediction to the algorithm. In the *No-Punishment* condition, where punishment is impossible, 59.3% delegated. The introduction of potential punishment thus significantly *reduces* delegation by 16.8 percentage points (Pearson Chi-squared, $p = 0.033$), opposing the direction posited by H1.[14] Figure 2 illustrates the difference. Delegation shares around 50% are consistent with subjects, on average, finding the claim of equal relative performance between themselves and the algorithm credible. Their behavior reflects neither systematic preference for one's own prediction nor systematic deference to the algorithm.

Result 1. *The possibility of punishment significantly reduces the likelihood that Player A delegates the final prediction to the algorithm.*

We next examine the robustness of the observed treatment difference in the delegation share and assess whether the treatment difference in delegation survives conditioning on socio-demographic characteristics and on Player A's actual and perceived task performance. Table 1 reports Probit estimates of the delegation decision.

Column (1) reports the unconditional treatment effect. Column (2) adds socio-demographic and other background characteristics. Columns (3) and (4) add Player A's actual performance (number of correct predictions in the first ten rounds) and her weighted-average belief about her own performance, respectively. Column (5) restricts the sample to subjects who passed the attention check.[15] The treatment effect is robust across all five specifications. In terms of average marginal effects, introducing the punishment possibility reduces the probability of delegation by between 16.8 and 19.4 percentage points (all $p \leq 0.031$).[16] Neither actual nor perceived performance is significantly associated with delegation in the pooled sample, although actual performance enters negatively and is weakly significant under the attention restriction ($p = 0.068$). We return to this pattern in discussing the mechanisms.

punishment comparisons use paired t-tests, with Wilcoxon signed-rank tests as non-parametric counterparts. Regressions report robust (HC1) standard errors. The pre-registration and any deviations from the preregistered analyses are described in Appendix C.

[14] A within-subject corroborator points in the same direction. Player As in the *No-Punishment* condition were asked, after their actual delegation decision, the hypothetical question of whether they would have delegated had punishment been possible. Among the 81 respondents, 12 would have stopped delegating under hypothetical punishment (25% of delegators) while only 5 would have started (15.2% of non-delegators), constituting weakly significant switching behavior towards making the prediction oneself with punishment possible (within-subject McNemar's test $p = 0.090$), and yielding a hypothetical delegation rate of 50.6%.

[15] While the attention check was included on another screen, it may still correlate with who took the delegation decision seriously.

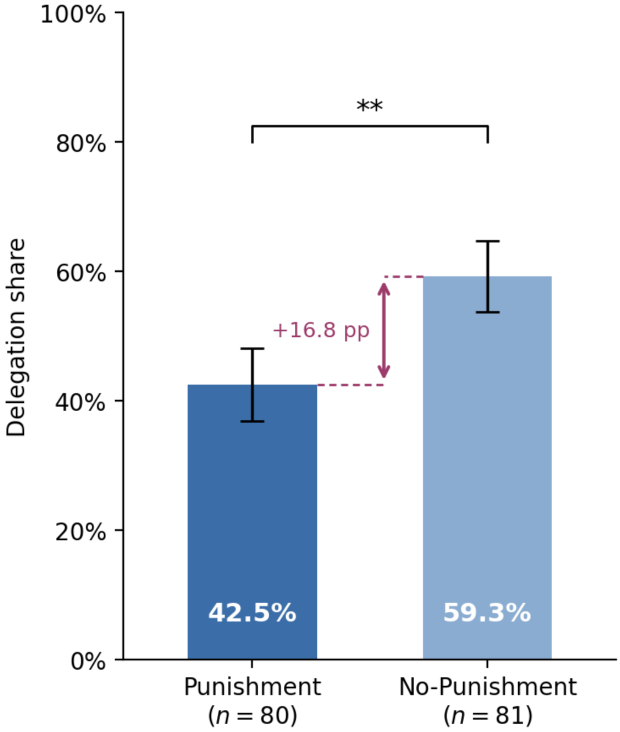
[16] Including a control for the random order in which the two delegation options appeared on screen leaves the treatment coefficient essentially unchanged (Punishment coefficient -0.48 ($p = 0.019$), with the "algorithm first" order indicator at 0.27 ($p = 0.21$), starting from the specification in Column (3)).

Table 1: Determinants of the delegation decision

	Dependent variable: *Delegation*				
	Full sample				*Passers*
	(1)	(2)	(3)	(4)	(5)
Punishment indicator	-0.423^{**}	-0.435^{**}	-0.472^{**}	-0.448^{**}	-0.517^{**}
	(0.199)	(0.204)	(0.206)	(0.205)	(0.230)
Task performance			-0.119		-0.158^{*}
			(0.079)		(0.086)
Perceived performance				-0.072	
				(0.056)	
Age		-0.002	-0.002	-0.002	0.002
		(0.009)	(0.009)	(0.009)	(0.010)
Female		0.175	0.222	0.212	0.407
		(0.227)	(0.228)	(0.229)	(0.254)
Socio-economic status		-0.038	-0.035	-0.031	-0.082
		(0.071)	(0.072)	(0.071)	(0.079)
Went to uni		0.036	0.059	0.025	0.124
		(0.227)	(0.225)	(0.227)	(0.247)
Technology score		-0.069	-0.048	-0.062	0.014
		(0.092)	(0.095)	(0.092)	(0.107)
Leadership position		0.404^{*}	0.422^{*}	0.399^{*}	0.253
		(0.225)	(0.226)	(0.225)	(0.253)
Constant	0.234^{*}	0.452	0.714	0.716	0.651
	(0.141)	(0.545)	(0.584)	(0.593)	(0.656)
AME of Punishment (pp)	-16.8^{**}	-16.8^{**}	-18.0^{**}	-17.2^{**}	-19.4^{**}
	$(p = 0.031)$	$(p = 0.030)$	$(p = 0.019)$	$(p = 0.026)$	$(p = 0.021)$
N	161	159	159	159	130
Pseudo R^2	0.020	0.039	0.049	0.047	0.064

Note: Coefficients from a Probit regression of the delegation decision on the Punishment indicator and a set of controls. Robust (HC1) standard errors in parentheses. Significance: $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$ (two-sided). Due to missing socio-demographic data, two subjects are dropped in Columns (2)–(5). The *AME of Punishment* reports the average change in the predicted probability of delegation from introducing the punishment possibility, in percentage points.

Figure 2: Delegation shares by condition



Note: Bars show the share of Player As who delegated the final prediction to the algorithm in each condition, with whiskers denoting one standard error of the mean. The treatment difference is significant at the 5% level under both a Pearson Chi-squared test ($p = 0.033$) and a Fisher's exact test ($p = 0.041$); $^{**}\, p < 0.05$.

Player B – Punishment behavior

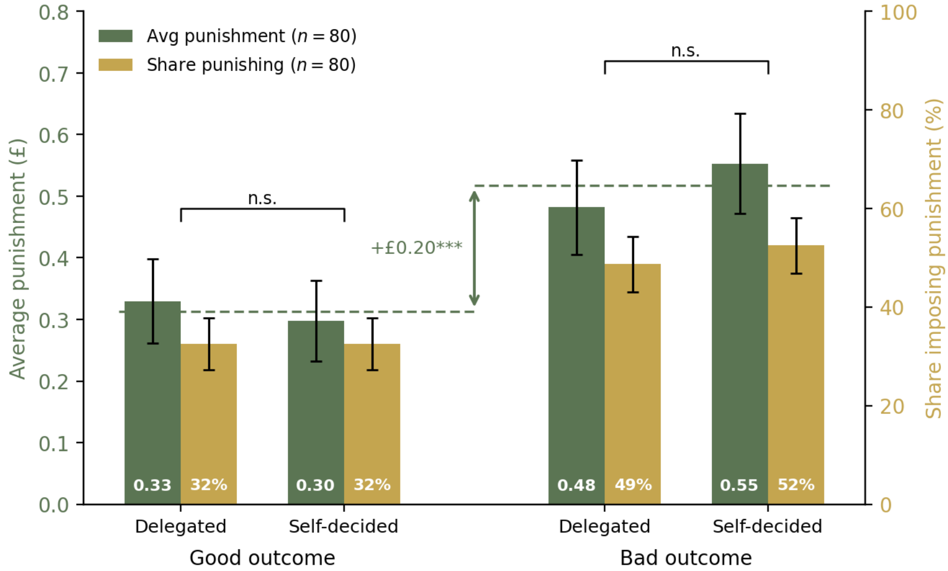
We now turn to Player B's punishment behavior in the *Punishment* condition, in which punishment was payoff-relevant. Punishment is substantial in all four (delegation, outcome) scenarios of the strategy method, with average deductions between £0.30 and £0.55 (Figure 3). Punishment responds strongly to the outcome. Player Bs punish bad outcomes significantly more than good outcomes, by £0.20 on average (paired t-test, $p < 0.001$), indicating that measured punishment carries real informational content rather than being random. The substantial punishment after *good* outcomes need not reflect noise either. Player Bs can hold views about how the final prediction should have been made, independently of the realized outcome, and can punish accordingly.[17]

Holding the outcome fixed, Player B punishes delegated and self-made decisions no differently. After a good outcome, average punishment is £0.33 for delegated and £0.30 for self-made decisions (paired t-test, $p = 0.53$). After a bad outcome, £0.48 versus £0.55. The punishment gap for bad outcomes thus runs in the direction predicted by H2 but is small and statistically insignificant (£0.07; paired t-test $p = 0.13$; Wilcoxon signed-rank $p = 0.54$).

Result 2. *Conditional on a low-payoff outcome, Player B does not punish delegated decisions*

[17]For example, Coffman (2011) likewise documents non-trivial punishment in settings where the decision-maker's action could not have been improved upon.

Figure 3: Player B punishment by delegation and outcome scenario



Note: For each (delegation, outcome) scenario, the green bars report the average punishment chosen by Player B (left y-axis, in pounds) and the gold bars the share of Player Bs imposing non-zero punishment (right y-axis, in percent), in the *Punishment* condition. Whiskers are ± 1 standard error of the mean. The brackets mark the within-outcome comparisons of average punishment between delegated and self-decided predictions (paired t-tests, $p = 0.53$ for good and $p = 0.13$ for bad outcomes). The dashed lines mark the mean of average punishment within each outcome group. The subsample of attention-check passers is shown in Appendix Figure 6. Hypothetical punishment in the *No-Punishment* condition is reported in Appendix Figure 8.

significantly less than self-made ones.

Regression analysis in Table 2 confirms this pattern. Punishment loads overwhelmingly on the outcome, while the effect of delegation for good outcomes is essentially zero. The *Delegated* $\times$ *Bad outcome* interaction, which captures any insulation from punishment after bad outcomes, amounts to -0.103 and is weakly significant ($p = 0.08$).[18] The conclusion is unchanged when we restrict the sample to subjects who passed the attention check, which was included on the punishment elicitation screen (Table 2, Column (2)).

[It is unclear whether we need the diff-in-diff table.]

The same pattern holds on the extensive margin.[19] For each outcome, the share of Player Bs imposing *any* punishment is indistinguishable between delegated and self-made decisions (Figure 3, right axis). Punishment is also far from universal. Nearly half of Player Bs (45.0%) impose no punishment in any of the four scenarios. Within each scenario, average punishment therefore reflects a punishing minority rather than light, diffuse sanctioning.

In sum, Player B punishes delegated decisions neither less harshly nor less frequently than

[18] In the regression, because punishment for the four scenarios is elicited within-subject, any stable characteristic of Player B is differenced out and cannot, by construction, confound the within-subject comparison. Thus, the control variables merely capture the between-subject association with Player B's *average* punishment level. We find no striking relationship between Player B's average punishment level and either their socio-demographic characteristics or their beliefs about Player A's *absolute* performance.

[19] The corresponding regression results can be found in Appendix Table 5.

Table 2: Determinants of Player B's chosen punishment

	Dependent variable: *Punishment (£)*	
	Full sample	*Passers*
	(1)	(2)
Delegated	0.032	0.001
	(0.052)	(0.054)
Bad outcome	0.256***	0.260***
	(0.071)	(0.080)
Delegated × Bad outcome	−0.103*	−0.078
	(0.058)	(0.052)
Player B belief about Player A perf.	−0.020	0.001
	(0.032)	(0.037)
Age	0.002	−0.001
	(0.005)	(0.006)
Female	−0.100	−0.064
	(0.128)	(0.140)
Socio-economic status	0.083*	0.072
	(0.047)	(0.049)
Went to uni	−0.196	−0.098
	(0.140)	(0.142)
Technology score	0.073	0.030
	(0.053)	(0.059)
Leadership position	0.083	0.061
	(0.142)	(0.164)
Constant	−0.077	0.020
	(0.328)	(0.374)
Observations	320	268
Subjects	80	67
Adj. R^2	0.077	0.030

Note: OLS regression of Player B's chosen punishment levels. The sample is stacked over the four (delegation, outcome) scenarios of the strategy method (4 observations per Player B). Column (1) uses the full Punishment-condition sample. Column (2) restricts to Player Bs who passed the attention check on the punishment-elicitation screen. Standard errors clustered at the Player B level.

self-made ones. In our setting, Player A can therefore not escape punishment by delegating the final prediction to the algorithm, but she also pays no punishment premium for doing so. Punishment responds almost entirely to the realized outcome.[20] This result also bears on *Result 1* as any treatment difference in delegation frequencies driven by punishment avoidance would be inconsistent with Player B's observed punishment behavior.

Mechanisms

Result 1 does not merely fail to support *Hypothesis 1*, but the effect is statistically significant in the opposite direction. The possibility of punishment discourages, rather than encourages, delegation to the algorithm. Thus, the prospect of punishment changes Player A's delegation decision, even though we detect no punishment penalty for delegating. In the remainder of this section, we examine the data more closely and weigh candidate explanations with the goal of reconciling the two main results.

Departures from canonical settings. *Hypothesis 1* drew on evidence from settings that differ from ours in ways that plausibly matter. In the modified dictator games of Bartling and Fischbacher (2012); Coffman (2011); Hamman et al. (2010); Oexl and Grossman (2013), the decision-maker chooses between a fair and an unfair allocation. Her intent is therefore transparent, the chosen allocation is implemented with certainty, and the unfair option benefits her directly at the recipient's expense. Punishment in these settings sanctions a deliberate, self-serving act, and delegation is attractive precisely because it blurs the link between the principal and that act. Our prediction task departs from these games. The outcome is uncertain, so a bad outcome may reflect bad luck rather than bad intent, and neither option affects Player A's own monetary payoff, leaving no self-serving choice to police. With little intent worth sanctioning, any intention-based punishment motive may lose its bite, and with it does the value to Player A of placing an algorithm between herself and the outcome. We nevertheless chose this task deliberately since prediction under genuine uncertainty, by a decision-maker with no direct material stake in the affected party's outcome, is the environment in which real-world algorithmic delegation typically operates. These departures, however, may explain the absence of the predicted effect, not its reversal. A reversal requires a motive that weighs against delegation when punishment is possible.

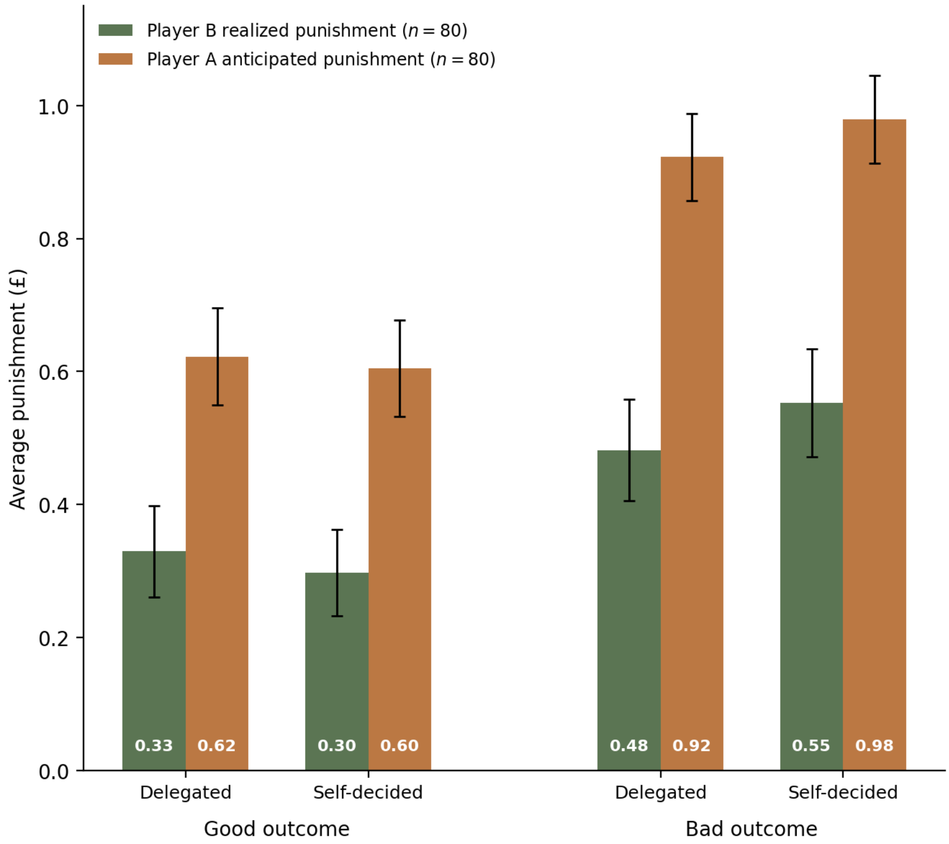
Anticipated differential punishment. The most natural candidate is anticipation of differential punishment. Player A may expect a delegated decision to be punished more harshly than a self-made one, and delegate less in response. Such an expectation would, for example, be consistent with previous evidence that observers view the delegation of morally consequential

[20] Hypothetical punishment choices elicited from Player Bs in the *No-Punishment* condition corroborate this conclusion (Appendix Figure 8). If anything, these Player Bs state that they would punish a delegated decision *more* than a self-made one after a bad outcome (£0.53 versus £0.43; paired t-test, $p = 0.069$).

decisions to machines relatively critically (e.g., Gogoll and Uhl, 2018; Bigman and Gray, 2018; Jauernig et al., 2022; Bonnefon et al., 2024).

After the delegation decision but before learning the outcome, we elicited Player A's beliefs about Player B's punishment in each of the four (delegation, outcome) scenarios. Anticipated punishment can affect behavior only where punishment is possible. We therefore focus the analysis of punishment beliefs on the *Punishment* condition. Two patterns emerge (Figure 4). First, Player As, on average, substantially overestimate punishment in every scenario. Second, Player As expect essentially the same punishment for delegated and self-made decisions, conditional on the outcome, correctly anticipating the pattern in Player B's actual punishment behavior (Result 2).[21] Thus, measured beliefs provide no support for a punishment anticipation mechanism for Result 1.[22]

Figure 4: Realized vs. anticipated punishment, by scenario



Note: Player B's realized punishment (green bars) and Player A's anticipated punishment (terracotta bars), in pounds, across the four (delegation, outcome) scenarios in the *Punishment* condition (full sample). Whiskers denote ±1 standard error of the mean. The subsample of attention-check passers is shown in Appendix Figure 7.

Player As, however, are far from unanimous in the punishment penalty they expect. We therefore ask whether these individual differences predict delegation. Table 3 reports Probit

[21]The expected punishment penalty for delegation is statistically indistinguishable from zero in both outcome states, and a within-subject test finds no asymmetry between the two.

[22]The same holds for the fully hypothetical beliefs elicited in the *No-Punishment* condition, which closely track those in the *Punishment* condition (Appendix Figure 8). The manipulation itself therefore appears not to have changed what subjects expected punishment to look like.

regressions of the delegation decision on Player A's belief difference in expected punishment (*no delegation* − *delegation*), so that higher values correspond to a larger expected punishment penalty for deciding herself. All specifications are restricted to the *Punishment* condition. Columns (1)–(3) use the simple average of the two outcome-conditional belief differences, Column (4) replaces the simple average with a weighted average, using Player A's elicited probability of producing the high payoff as the weight, and Column (5) enters the two differences separately.

Table 3: Delegation and beliefs about punishment

	Dependent variable: *Delegation*				
	Full Punishment		*Punishment, passers*		
	(1)	(2)	(3)	(4)	(5)
Belief difference, average ($\overline{\Delta}$)	0.346 (0.407)	0.398 (0.448)	1.661** (0.700)		
Belief difference, weighted by Pr(good) (Δ^w)				1.992** (0.794)	
Belief difference, bad outcome (Δ^{bad})					0.590 (0.424)
Belief difference, good outcome (Δ^{good})					0.961** (0.412)
Task performance		−0.208* (0.111)	−0.362*** (0.136)	−0.342** (0.142)	−0.381*** (0.138)
Age		0.023* (0.012)	0.039** (0.016)	0.040** (0.016)	0.039** (0.016)
Female		0.152 (0.350)	0.950** (0.427)	1.026** (0.406)	0.845* (0.444)
Socio-economic status		−0.049 (0.111)	−0.006 (0.131)	−0.055 (0.133)	0.001 (0.126)
Went to uni		0.071 (0.363)	−0.278 (0.412)	−0.169 (0.408)	−0.275 (0.409)
Technology score		−0.061 (0.138)	0.048 (0.170)	0.072 (0.173)	0.033 (0.167)
Leadership position		0.302 (0.312)	−0.301 (0.408)	−0.279 (0.408)	−0.245 (0.411)
Constant	−0.198 (0.142)	−0.345 (0.826)	−1.228 (1.155)	−1.281 (1.147)	−1.115 (1.165)
AME of belief measure (pp per £0.10)	1.3 ($p = 0.388$)	1.4 ($p = 0.370$)	4.7** ($p = 0.011$)	5.5*** ($p = 0.006$)	
AME: good-outcome difference (pp per £0.10)					2.7** ($p = 0.010$)
AME: bad-outcome difference (pp per £0.10)					1.7 ($p = 0.165$)
Controls	×	✓	✓	✓	✓
Passers only	×	×	✓	✓	✓
N	80	78	60	60	60
Pseudo R^2	0.007	0.098	0.248	0.263	0.252

Note: Probit estimates with robust (HC1) standard errors in parentheses. Sample restricted to the *Punishment* condition throughout. Belief differences are within-subject *(no delegation)* − *(delegation)* in expected punishment, so positive coefficients indicate that subjects who expect to be punished more harshly for not delegating (relative to delegating) are more likely to delegate. Columns (1)–(3) use the simple average of the good-outcome and the bad-outcome belief difference. Column (4) instead weights the good-outcome difference by Player A's elicited probability of producing the high payoff and the bad-outcome difference by the complementary probability. Column (5) enters the two differences separately. The *AME* rows report the average marginal effect of the belief measure entered in the respective column on the probability of delegation, in percentage points per £0.10 increase in the corresponding belief difference.

Expected differential punishment does not predict delegation in the full sample (Columns (1)–(2)). Among subjects who passed the attention check on the belief-elicitation screen, the relationship sharpens. A £0.10 larger expected punishment penalty for deciding herself is associated with a 4.7 percentage-point higher probability of delegation (Column (3), $p = 0.011$), and with 5.5 percentage points under the weighted specification (Column (4), $p = 0.006$).[23]

Taken together, the belief evidence does not support anticipated differential punishment as the mechanism behind Result 1. For anticipation of punishment to generate a 17 percentage-point delegation gap between conditions, Player As would need to expect systematically harsher punishment for delegated decisions. The measured beliefs show no such expectation, and such an expectation would not have been correct, as Player B's realized punishment exhibits no delegation penalty either (Result 2). What the belief evidence does show is heterogeneity. Within the *Punishment* condition, Player As whose beliefs embody a larger penalty for deciding themselves are more likely to delegate. Anticipated punishment is thus a dimension along which subjects differ, but it provides no aggregate push toward less delegation under punishment. We nonetheless read both findings with caution. First, hypotheticality may introduce noise as only two of the four scenarios could still materialize after Player A's delegation decision in the *Punishment* condition.[24] Second, the elicitation of beliefs was unincentivized and took place after the delegation decision, so reports may reflect post-hoc rationalization rather than the beliefs that drove the choice.

Result 3. *On average, Player A expects no punishment penalty for delegation, mirroring Player B's actual punishment behavior. Within the Punishment condition, however, Player As who expect a larger penalty for deciding themselves are more likely to delegate.*

Sample composition. Result 1 is not an artifact of sample composition either. Treatment randomization produced balance across socio-demographic and other background characteristics (Appendix A.1, Table 4). The same holds for performance. Average performance in the first ten rounds is balanced across conditions (2.76 vs. 3.09 correct predictions out of ten in the *Punishment* and *No-Punishment* conditions, respectively; Figure 5).[25] Player As in both conditions also exhibit similar overconfidence about their own performance (4.65 vs. 4.87, respectively).[26] Finally, the treatment effect is robust to controlling for actual and perceived performance (Table 1, Columns (3)–(4)).

[23] In the decomposition of Column (5), the good-outcome belief difference carries the relationship (2.7 percentage points per £0.10, $p = 0.010$), while the bad-outcome difference is positive but noisier (1.7 percentage points, $p = 0.165$). The two differences are nearly orthogonal in this sample, so we read their coefficients as additive partial slopes rather than competitors for the same variance. Under expected-utility reasoning, however, only their probability-weighted sum should enter the decision rule. We therefore treat Column (5) merely as a descriptive decomposition.

[24] In the *No-Punishment* condition all four scenarios were purely hypothetical.

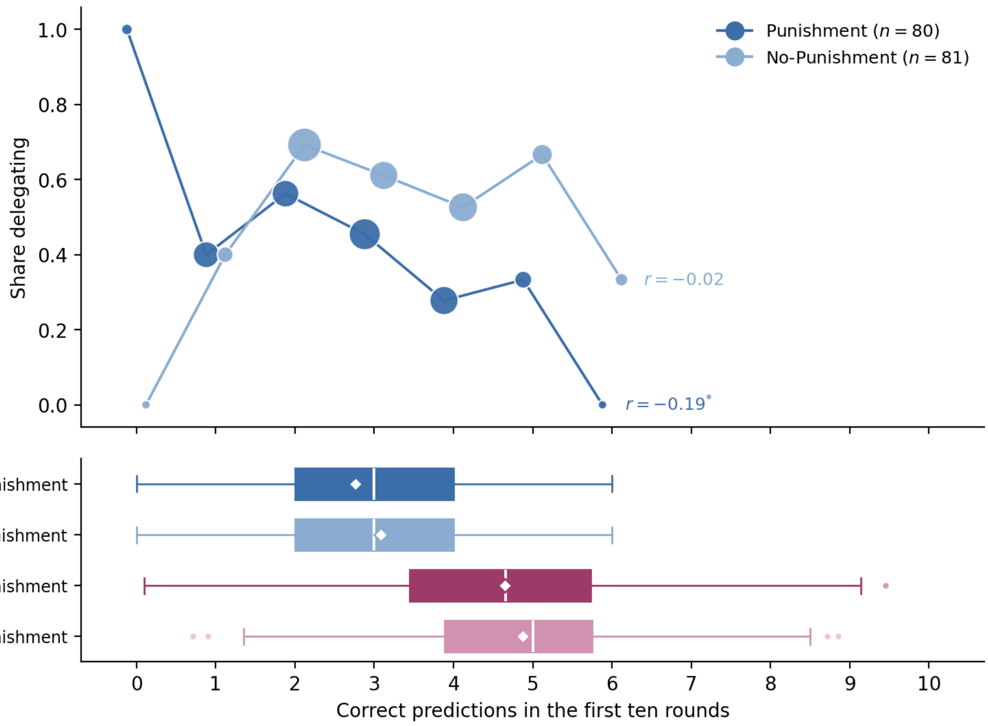
[25] To the extent this small raw difference matters at all, it works against Result 1, because Player A in the *Punishment* condition performs slightly worse, and as we will see, it is worse performers who tend to delegate more, so imbalance would attenuate rather than generate the treatment difference.

[26] Confidence is measured by the weighted-average belief about own performance. This overconfidence may partly reflect the information that the algorithm performs equally well.

Performance and delegation. What differs across conditions is *who* delegates. Table 1 provided a first indication that task performance is negatively associated with the decision to delegate across conditions (e.g., Column (5)). This negative relationship is considerably stronger in the *Punishment* condition (Table 3) than in the *No-Punishment* condition (Figure 5). For perceived performance, a directionally similar but much weaker pattern emerges in the *Punishment* condition.[27] [We should probably never make such claims without showing the corresponding regression results?] Actual performance thus discriminates sharply between delegators and non-delegators, whereas perceived performance does not. Delegators under punishment are not subjects who *believe* they perform poorly but subjects who *do*. As a result, the delegation gap between conditions therefore widens with performance. Player As who delegated in the *Punishment* condition scored on average 2.47 out of 10 in the first ten rounds, compared with 3.06 for delegators in the *No-Punishment* condition (t-test, $p = 0.039$). It is precisely the better performers who, when punishment is on the table, keep the decision for themselves.

Result 4. *In the Punishment condition, better-performing Player As are less likely to delegate. The additional delegation in the No-Punishment condition comes disproportionately from better performers.*

Figure 5: Delegation, performance, and confidence by condition



Note: The top panel shows the delegation share of Player A at each score, by treatment. Dot size is proportional to the number of subjects at that score. The bottom panel shows the distributions of actual and perceived (the weighted-average belief about own performance) scores, by condition. Boxes span the interquartile range, the white line marks the median, and the white diamond the mean.

Our design rules out the natural explanation for this relationship, namely that worse perform-

[27] Player A's weighted-average belief about her own performance would enter the regressions with a small negative coefficient but is never statistically significant.

ers delegate because they believe the algorithm outperforms them. In Feier et al. (2022), the algorithm's performance distribution is matched to the distribution of human performance at the population level. A high-performing subject who recognizes her position in that distribution therefore rationally expects to outperform the algorithm, and a low-performing subject rationally expects the reverse. Delegation that declines with performance is then exactly what expected-payoff maximization predicts. In our experiment, by contrast, performance is equalized between the algorithm and each *individual* subject, and this is common knowledge. The pattern in the *Punishment* condition must therefore reflect something other than beliefs about relative ability.[28]

One reading of the relationship between performance and delegation invokes a preference over *process ownership* of the outcome. Player As with a good chance of producing the high payoff may prefer to bring it about themselves rather than to have the algorithm do so. Conversely, Player As with little chance of success may delegate to avoid being the proximate cause of a likely failure. While this motive can rationalize a negative relationship between performance and delegation, it would predict that relationship in both conditions and no difference in delegation rates between them, unless the preference is activated by accountability, only becoming behaviorally consequential once Player B holds an enforcable stake in the decision rather than merely a material one. Relatedly, Player A's preferences for delegation could vary with the amount of uncertainty inherent in a decision. Dietvorst and Bharti (2020) document that the preference for algorithms declines as such uncertainty rises. In our task, where most subjects score below 5 out of 10, better performance moves the success probability toward 0.5, arguably the point of maximal outcome uncertainty. A distaste for delegating uncertain decisions would then depress delegation among better performers. However, like process ownership, this channel operates equally in both conditions and cannot, on its own, explain the difference between them. Comment: How does uncertainty in a task relate to difficulty of a task?

Effort to outperform the algorithm. Lastly, the prospect of punishment could motivate Player A to exert additional effort when making the final prediction for Player B. Although the algorithm's chance of success is fixed by Player A's past performance, the final prediction itself has yet to be made. Player A may thus believe that, by trying especially hard, she can outperform the algorithm and thereby give Player B a better chance of the high payoff. To the extent that she expects milder punishment after a good outcome, the prospect of punishment would then push her toward making the prediction herself. This account yields a testable prediction. Relative to the first ten rounds, non-delegators should improve their performance more in the *Punishment* condition than in the *No-Punishment* condition. Similar improvement in

[28]We cannot rule out that some subjects did not fully internalize the individual-level equalization. Two observations mitigate this concern. First, any such misunderstanding should arise in both conditions alike, so it can explain neither why the relationship between performance and delegation is markedly stronger under punishment nor why delegation rates differ between conditions. Second, under this alternative, delegation should decline with perceived own performance, with more confident subjects retaining the decision. In the data, perceived performance is statistically indistinguishable between delegators and non-delegators.

both conditions would instead indicate a punishment-independent motive, for instance altruism toward Player B, which cannot explain Result 1.

The data do not lend support to this story. Delegators do not make a final prediction themselves, so effort on the final prediction is observable merely for non-delegators. Measuring improvement as the difference between a subject's average absolute prediction error over the first ten rounds and her absolute error on the final prediction, non-delegators in the *No-Punishment* condition improve on their record, whereas non-delegators in the *Punishment* condition do not. Because all Player As faced the same images in the first 10 rounds and on the final prediction, this differential cannot reflect task-specific variation. Realized outcomes tell the same story with 45.5% of non-delegators in the *No-Punishment* condition earning the high payoff for Player B, compared with 21.7% in the *Punishment* condition, even though the two groups' performances are statistically indistinguishable over the first ten rounds (3.12 vs. 2.98 correct predictions; $p = 0.66$). If anything, effort seems to rise where punishment is impossible, which is the opposite of what a punishment-motivated effort story would predict.

The data do not support this story. Delegators do not make a final prediction themselves, so effort on the final prediction is observable only for non-delegators. We measure improvement as the difference between a subject's average absolute prediction error over the first ten rounds and her absolute error on the final prediction. By this measure, non-delegators in the *No-Punishment* condition improve, whereas non-delegators in the *Punishment* condition do not. Because all Player As faced the same images in the first ten rounds and on the final prediction, this contrast between conditions cannot reflect variation in image difficulty. Realized outcomes tell the same story. Among non-delegators, 45.5% in the *No-Punishment* condition earn the high payoff for Player B, compared with 21.7% in the *Punishment* condition, even though the two groups' performance over the first ten rounds is statistically indistinguishable (3.12 vs. 2.98 correct predictions; t-test, $p = 0.66$). How do we reconcile balance in performance across the first ten tasks between treatements, the fact that only in the Punishment conditions better player As do not delegate and that there is balance among non-delegators between treatments? Answer: The reconciliation of your three facts is then a clean decomposition, and it answers your original question properly. Non-delegators exceed their own condition's mean by 0.22 under Punishment but only 0.04 under No-Punishment, so among non-delegators the pool gap and the selection differential offset, 0.32 - 0.18 = 0.14, which is your insignificant comparison. Among delegators the shortfalls are 0.29 and 0.02, so the two forces compound, 0.32 + 0.27 = 0.59, which is your significant one. In words, Punishment non-delegators look similar to No-Punishment non-delegators not because selection is absent but because they are more strongly positively selected from a somewhat weaker pool. The delegator comparison is where pool and selection push the same way, which is exactly why it is the statistically informative cut. If anything, performance on the final prediction is higher where punishment is unavailable, the opposite of what a punishment-motivated effort account predicts.[29]

[29]Screen time points in the same direction. In the *Punishment* condition Player As spend less time on the final prediction screen than in the *No-Punishment* condition (medians 16 vs. 22 seconds; Mann-Whitney $p = 0.019$).

Result 5. *The prospect of punishment does not induce additional effort. Non-delegators improve on their performance less in the Punishment condition than in the No-Punishment condition.*

The signaling value of not delegating. Although the prospect of punishment does not appear to induce additional effort (Result 5), the delegation choice itself may communicate effort. By keeping the decision, Player A may want to convey to Player B that she takes the task seriously and intends to try hard on his behalf. Delegating, by contrast, could be perceived as shirking a decision that matters to another person. The signal lies in the choice itself rather than in any effort that follows, so Result 5 does not speak against this account. Result 3, however, seems to. The signal has monetary value only if Player A expects delegation, as apparent shirking, to draw harsher punishment, whereas in the *No Punishment* condition, Player A could shirk without monetary consequences. Result 3 indicates that Player As in the *Punishment* condition, on average, hold no such expectation either, which leaves little monetary gain from the signal.[30]

This tension can be resolved in two ways. First, Player A may care about Player B's judgment itself, beyond any punishment attached to it. For this concern to explain the treatment difference, however, it would have to be activated by the prospect of punishment itself, even though the expected amount of punishment does not differ by delegation choice. Second, as discussed above, our belief measure was unincentivized and elicited only after the delegation decision, so it may capture the beliefs behind that decision imperfectly. Either way, a concern about being perceived as shirking would weigh against delegation precisely when punishment is possible. In our environment, where no self-serving option exists and outcomes are noisy, the delegation choice is the only deliberate act on which Player B can condition punishment, a fact that may lend this perception particular weight.

Taking stock. The reversal in Result 1 is robust, but its mechanism remains unresolved. It is not a composition artifact. The conditions are balanced in observables, performance, and confidence, and the treatment effect survives the corresponding controls. Belief evidence suggests that anticipated differential punishment cannot explain the reversal (Result 3). On average, Player A expects no punishment penalty for delegation, mirroring Player B's realized punishment (Result 2), although Player As who expect a larger penalty for deciding themselves are more likely to delegate. Punishment-induced effort cannot explain the reversal either, because non-delegators under punishment spend less time on the prediction for Player B and improve less, not more (Result 5). What the data do show is selection. It is the better-performing

The same holds within subject. Non-delegators in both conditions spend more time on the final prediction than on an average round of the initial prediction task, but the increase is smaller in the *Punishment* condition (+5.0 vs. +8.2 seconds; Mann-Whitney $p = 0.058$).

[30] The heterogeneity in Result 3 leaves room for this channel at the individual level. Player As who expect a larger punishment penalty for delegating are more likely to keep the decision (Table 3). An expected penalty of zero on average, however, provides no aggregate push toward less delegation under punishment.

Player As who retain the decision when punishment is possible (Result 4). Preferences over process ownership or outcome uncertainty are consistent with this selection but operate in both conditions alike. They could explain Result 1 only if accountability itself activated them, a hypothesis our design cannot test because the possibility of punishment and the presence of accountability coincide by construction. The signaling value of not delegating requires no actual additional effort (Result 5) and survives the belief evidence if Player A cares about Player B's judgment itself (Result 3), but, similarly, rests on being activated by the prospect of punishment itself.

5 Conclusion

tbd

Our findings add a potential determinant of algorithm use to the drivers documented in the aversion literature. Existing accounts locate the reluctance to rely on algorithms in the algorithm or the task, in doubts about ability, discomfort with opacity, or a taste for decisions made by hand. In our setting, the algorithm's performance is equalized with the decision-maker's by construction, and delegation nevertheless declines once the affected party can hold the decision-maker accountable. What appears to deter delegation is not what the algorithm does, but what delegating may say about the person who chooses it. Keeping the decision can signal engagement on behalf of the affected party, whereas handing it to the algorithm could be perceived as shirking. Accountability structures may thus shape algorithm use through perception concerns rather than through beliefs about performance. For the design of such structures, this cuts both ways. Holding human users accountable keeps them engaged with consequential decisions, in line with the intention of human-oversight requirements. Yet it may also discourage delegation in situations where the algorithm would decide just as well, and it does so disproportionately among the most capable decision-makers.

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Appendix

A Additional Tables

A.1 Condition balance — Player A

Table 4: Condition balance — Player A

Variable	*Punishment*	*No-Punishment*	p-value
Age	39.026	37.877	0.552
Female	0.487	0.494	1.000
Socio-economic status	5.179	5.321	0.564
Went to uni	0.575	0.556	0.928
Technology score	2.087	2.284	0.322
Leadership position	0.362	0.333	0.824

Note: Mean values by condition for Player A, with p-values from two-sided t-tests for continuous variables and Chi-squared tests for binary variables. *Socio-economic status* is self-reported on a 1–10 scale. *Went to uni* indicates a university degree. *Technology score* is constructed from weekly device usage, programming skills, cryptocurrency knowledge, technology use at work, and similar items. *Leadership position* indicates self-reported management or supervisory experience.

Complementary to the treatment comparisons of individual observable characteristics in Table 4, we can test their joint significance by regressing the treatment indicator on all characteristics in a linear probability model. The observables are jointly uninformative about treatment ($F(6, 152) = 0.36$, $p = 0.90$; $R^2 = 0.01$; $N = 159$). A probit specification yields the same conclusion (likelihood-ratio $\chi^2(6) = 2.27$, $p = 0.89$).

We do not separately report a balance table for Player B. Player B's punishment behavior is identified within-subject across the four (delegation, outcome) scenarios of the strategy method, so balance across conditions is not strictly required for the punishment analysis.

A.2 Extensive margin of punishment

Table 5 re-estimates the punishment regression on the *extensive* margin, replacing the punishment amount with an indicator for whether Player B imposed any punishment in a given scenario, estimated by probit. The conclusion matches the intensive-margin result in the body. A bad outcome sharply raises the probability of punishment—an average marginal effect of about 18 percentage points ($p < 0.001$)—whereas delegation leaves it essentially unchanged: the average marginal effect of delegation is -1.8 percentage points and statistically insignificant ($p = 0.46$; -2.9 percentage points among attention-check passers, $p = 0.29$). Player A is therefore no less *likely* to be punished when she delegates, just as she is punished no less *harshly*.

Table 5: Extensive margin of punishment — probit for Pr(punishment > 0)

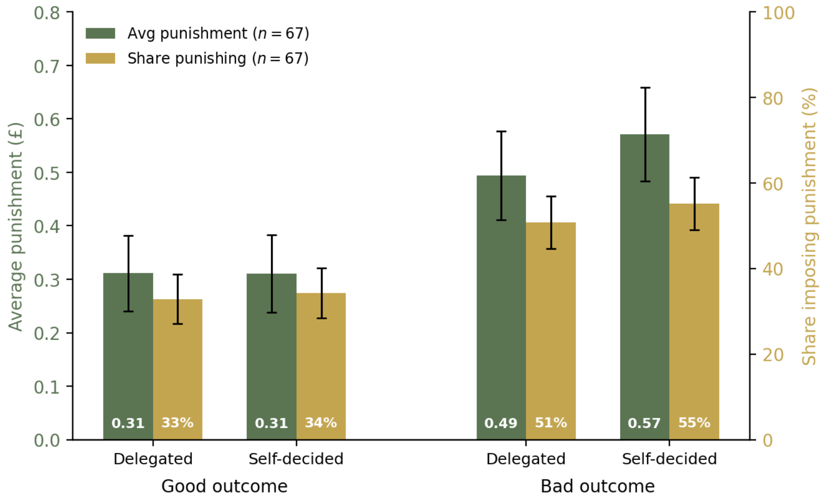
	Dependent variable: *Any punishment*	
	Full sample	*Passers*
	(1)	(2)
Delegated	0.002	−0.042
	(0.077)	(0.078)
Bad outcome	0.556***	0.563***
	(0.139)	(0.153)
Delegated × Bad outcome	−0.099	−0.072
	(0.095)	(0.103)
Player B belief about Player A perf.	−0.083	−0.030
	(0.075)	(0.082)
Age	−0.006	−0.013
	(0.013)	(0.015)
Female	0.115	0.230
	(0.272)	(0.291)
Socio-economic status	0.160	0.140
	(0.126)	(0.129)
Went to uni	−0.827***	−0.706**
	(0.320)	(0.349)
Technology score	0.062	−0.023
	(0.118)	(0.131)
Leadership position	0.062	−0.071
	(0.293)	(0.321)
Constant	−0.382	−0.182
	(0.807)	(0.864)
AME of Delegated (pp)	−1.8	−2.9
	($p = 0.462$)	($p = 0.285$)
AME of Bad outcome (pp)	18.1***	19.4***
	($p < 0.001$)	($p < 0.001$)
Observations	320	268
Subjects	80	67
Pseudo R^2	0.096	0.081

Note: Probit regression of an indicator for non-zero punishment ($1\{\text{punishment} > 0\}$) on a delegation indicator, a bad-outcome indicator, their interaction, Player B's belief about Player A's performance, and socio-demographic controls; the sample is stacked over the four (delegation, outcome) cells of the strategy method (4 observations per Player B). Column (1) uses the full Punishment-condition sample; Column (2) restricts to attention-check passers. Standard errors clustered at the Player B level. The *AME* rows report discrete average marginal effects on the probability of imposing any punishment, in percentage points (for Delegated, the change from $0 \to 1$ with the interaction moved consistently; for Bad outcome, likewise), computed by counterfactual prediction with p-values from a seeded 400-replication subject-cluster bootstrap. Significance: $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$ (two-sided).

B Additional figures

B.1 Punishment among attention-check passers

Figure 6: Player B punishment by delegation and outcome scenario, attention-check passers

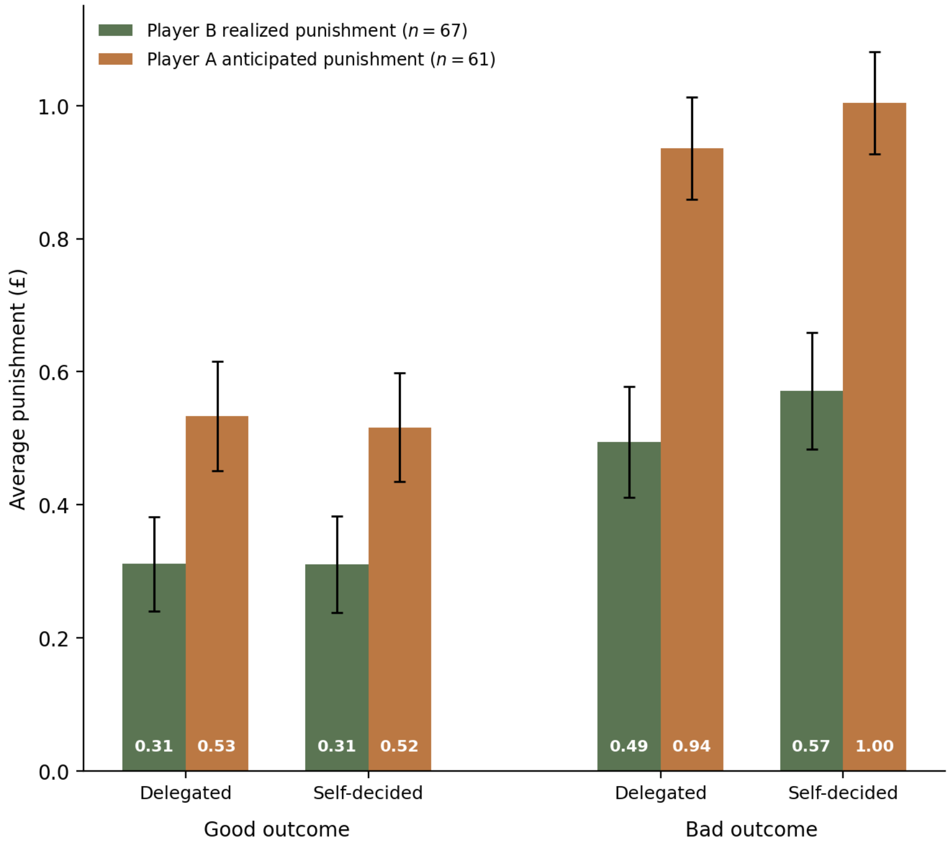


Note: As in Figure 3, but restricted to Player Bs who passed the attention check on the punishment-elicitation screen ($n = 67$). The four (delegation, outcome) scenarios of the strategy method are grouped by outcome. For each scenario, the green bars report the average punishment (left y-axis, in pounds; whiskers are ± 1 standard error of the mean) and the gold bars the share of Player Bs imposing non-zero punishment (right y-axis, in percent; whiskers are ± 1 standard error of the share). Values are printed at the base of each bar. The left-axis scale matches Figure 3.

B.2 Hypothetical punishment

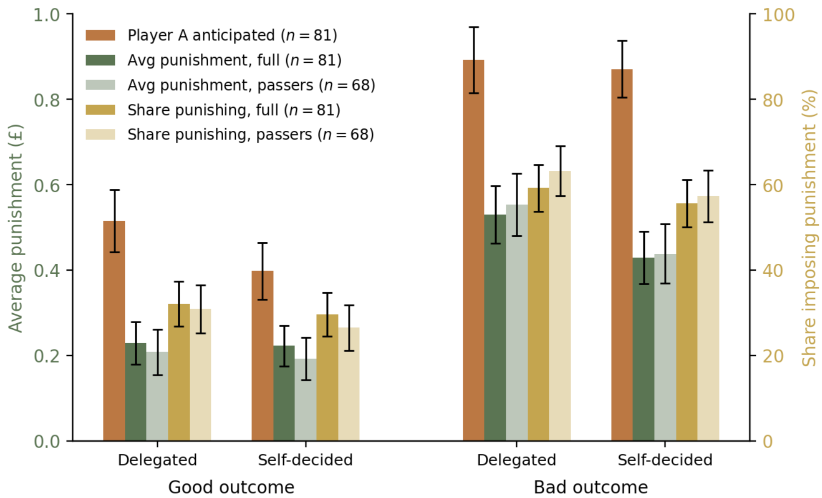
Hypothetical punishment in the *No-Punishment* condition follows the same outcome-driven pattern as actual punishment (Figure 8). After good outcomes, stated punishment is nearly identical for delegated and self-made decisions (£0.23 versus £0.22; paired t-test, $p = 0.91$). After bad outcomes, the comparison reverses relative to the direction predicted by *Hypothesis 2*. Player Bs state that they would punish a delegated decision more than a self-made one (£0.53 versus £0.43; paired t-test, $p = 0.069$), a weakly significant difference. Even in stated terms, delegation thus offers Player A no insulation from punishment. Player A's anticipated punishment, shown alongside, overstates hypothetical punishment in every scenario, mirroring the overestimation of actual punishment in the *Punishment* condition (Figure 4).

Figure 7: Realized vs. anticipated punishment, attention-check passers



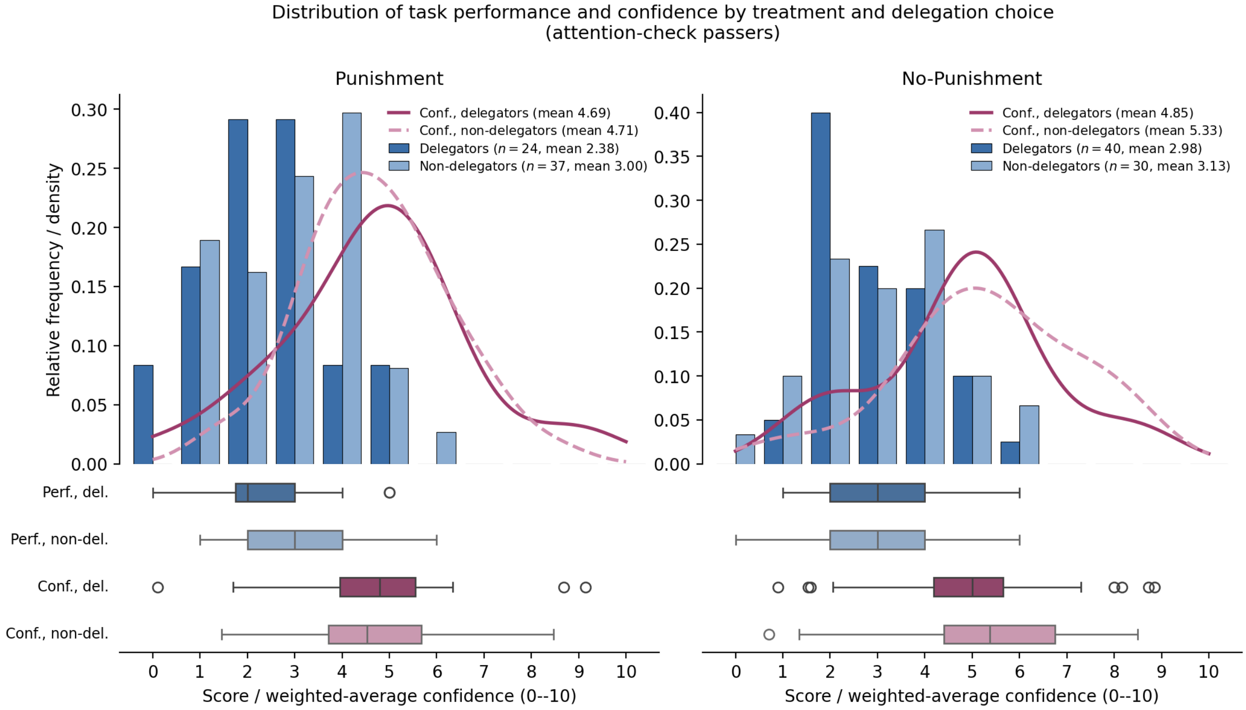
Note: As in Figure 4, but restricted to attention-check passers (Player A $n = 61$, Player B $n = 67$).

Figure 8: Hypothetical punishment by delegation and outcome scenario (*No-Punishment* condition)



Note: As in Figure 3, but for the *No-Punishment* condition (full sample $n = 81$, passers $n = 68$), in which Player Bs answered the same punishment questions but could not impose punishment in practice. The four (delegation, outcome) scenarios are grouped by outcome. For each scenario, the green bars report the average hypothetical punishment (left y-axis, in pounds; whiskers are ± 1 standard error of the mean) and the gold bars the share of Player Bs choosing non-zero punishment (right y-axis, in percent; whiskers are ± 1 standard error of the share); the full sample is shown at full opacity and the attention-check passers at lower opacity. In addition, the terracotta bars report Player A's anticipated punishment in the *No-Punishment* condition (hypothetical, full sample, $n = 81$).

Figure 9: Performance and confidence distributions by treatment and delegation choice



Note: Restricted to attention-check passers. Within each treatment, the top panel shows the share of Player As at each first-ten-rounds score, separately for those who delegated (dark blue) and those who did not (light blue), with the weighted-average performance belief overlaid as a KDE (delegators solid magenta, non-delegators dashed pink). Boxplot strips below the top panel report the same four series. In the *Punishment* condition, actual performance discriminates between delegators and non-delegators (means 2.38 vs. 3.00) while perceived performance does not (means 4.69 vs. 4.71). In the *No-Punishment* condition, neither actual nor perceived performance discriminates (actual: 2.98 vs. 3.13; perceived: 4.85 vs. 5.33). The additional delegators in *No-Punishment* therefore come disproportionately from the middle and upper portion of the actual performance distribution.

Figure 10: Experimental timeline — Player A

To be added.

Figure 11: Experimental timeline — Player B

To be added.

B.3 Performance and confidence by delegation choice

B.4 Experimental timelines

C Pre-registration

The study was pre-registered on AsPredicted under the title "Algorithms and Responsibility" (#133980) on May 30, 2023, after the collection of a pilot sample and before the collection of the main sample. The pre-registration is available at https://aspredicted.org/hs9az8.pdf and is reproduced in full on the following page.

The analyses in the paper deviate from this document in three respects. First, we recruited 322 participants instead of the planned 320. Attrition during the experiment required re-matching and produced one additional pair in the *No-Punishment* condition. Second, we compare delegation rates across conditions using a Pearson chi-squared test and Fisher's exact test rather than the preregistered t-test. The delegation decision is binary, so these tests are the appropriate choice, and the conclusion is unchanged under either procedure. Third, the pre-registration specifies Mann-Whitney U and Kolmogorov-Smirnov tests for the punishment comparisons. Both are tests for independent samples, whereas punishment is elicited within-subject via the strategy method. We therefore report the Wilcoxon signed-rank test as the within-subject non-parametric counterpart alongside the parametric tests. Beyond these deviations, the pre-registration left open whether participants who failed the attention checks would be excluded. We report all analyses for the full sample in the main text and repeat them for the subsample of attention-check passers in the appendix.

Algorithms and Responsibility (#133980)

Author(s)
This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:
2023/05/30 16:27 (PT)

1) Have any data been collected for this study already?
It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?
We test whether:
RQ 1. punishment for a decision can be avoided if it is delegated to an algorithm (conditional on performance)
RQ 2. a punishment motive enters the delegation decision.

3) Describe the key dependent variable(s) specifying how they will be measured.
The experiment features a delegator and an evaluator. The delegator has to make a decision that is payoff-relevant only to the evaluator. The delegator may decide herself or delegate the decision to an algorithm, which is calibrated to match the delegators performance. Depending on the treatment, the evaluator may punish the delegator conditional on the outcome of the decision and the delegation decision (strategy method).
For RQ 1, we will look at evaluator punishment behavior and compare the amount and frequency of punishment between a delegated and non-delegated decision conditional on the outcome.
For RQ 2, we will look at the delegation frequency between the treatments (punishment possible vs punishment not possible).

4) How many and which conditions will participants be assigned to?
There are two conditions (punishment possible/no punishment possible) and two roles in each condition (delegator/evaluator).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
I will run parametric (t-test) and non-parametric (MWU, Kolmogorov-Smirnov) tests between punishment (amount and frequency) for delegated and non-delegated outcomes, conditional on the outcome. Similarly, I will run linear regressions with punishment behavior as the dependent variable and explore whether expected performance or any socio-demographic characteristics are predictive of differential punishment.
For delegators, I will again, run parametric (t-test) tests on delegation choices between treatments. Moreover, I will employ Probit regressions with the dependent variable being the delegation decision and a set of explanatory variables including treatment, beliefs about performance and socio-demographic characteristics.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
I will exclude people that access the experiment via mobile device. Moreover, I may exclude participants that failed the attention checks.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.
I will recruit subjects via Prolific. I plan to recruit 320 participants, 160 per treatment, 80 per role per treatment.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)
I already ran a pilot with 25 observations in the punishment treatment. I plan to include the data in the analysis.

Secondary analysis will include an investigation on how performance and beliefs about performance impact results. Moreover, I will look at the delegator's beliefs about expected punishment and their consistency with the delegation decision.

At this point, I already want to state that I may run another treatment in which the delegator may either delegate to another participant or an algorithm.